

Parameters

Parameter	Description	Value
outer-header	Specifies load balancing based on the outer header.	-
inner-header	Specifies load balancing based on the inner header.	-

Views

Load balancing profile view

Default Level

2: Configuration level

Usage Guidelines

Usage Scenario

For tunnel packets, such as GRE and VXLAN packets, you can run the **tunnel** command to configure hash calculation based on the outer or inner header in the specified load balancing profile.

Precautions

For tunnel packets, route selection involving inner packet factors takes effect only in known unicast scenarios.

Example

Configure load balancing for tunnel packets based on inner-header in a load balancing profile.

```
<HUAWEI> system-view
```

```
[~HUAWEI] load-balance profile abc
```

```
Warning: The profile default has been used, and this command will rename it. Continue? [Y/N]:y
```

```
[*HUAWEI-load-balance-profile-abc] tunnel inner-header
```

2.4.3 VLAN Configuration Commands

2.4.3.1 bandwidth (VLANIF interface view)

Function

The **bandwidth** command sets the bandwidth of a VLANIF interface.

The **undo bandwidth** command deletes the configured bandwidth of a VLANIF interface.

By default, the bandwidth of a VLANIF interface is 1000 Mbits/s.

Format

bandwidth *bandwidth*

undo bandwidth

Parameters

Parameter	Description	Value
<i>bandwidth</i>	Specifies the bandwidth of a VLANIF interface.	The value ranges from 1 to 1000000, in Mbits/s.

Views

VLANIF interface view

Default Level

2: Configuration level

Usage Guidelines

Usage Scenario

The bandwidth command mainly ensures that the network management station (NMS) can acquire the bandwidth of the VLANIF interface. The NMS can check the interface bandwidth through the two objects ifSpeed and ifHighSpeed in IF-MIB.

- If the configured bandwidth is smaller than 4000 Mbit/s, ifSpeed and ifHighSpeed are respectively displayed as bandwidth x 1000 x 1000 and bandwidth.
- If the configured bandwidth is equal to or larger than 4000 Mbit/s, ifSpeed and ifHighSpeed are respectively displayed as 4294967295 (0xFFFFFFFF) and bandwidth.

Example

```
# Set the bandwidth of VLANIF 2 to 10000 Mbits/s.
<HUAWEI> system-view
[~HUAWEI] vlan 2
[*HUAWEI-vlan2] quit
[*HUAWEI] interface vlanif 2
[*HUAWEI-Vlanif2] bandwidth 10000
```

2.4.3.2 damping time (VLANIF interface view)

Function

The **damping time** command enables VLAN damping and sets a delay time after which a VLANIF interface goes Down.

The **undo damping time** command restores the default setting.

By default, the VLAN damping function is disabled, and the delay time is 0 seconds, which means that the VLANIF interface goes Down immediately after the VLAN goes Down.

Format

damping time *delay-time*

undo damping time

Parameters

Parameter	Description	Value
time <i>delay-time</i>	Specifies a delay time after which a VLANIF interface goes Down.	The value is an integer ranging from 0 to 20, in seconds.

Views

VLANIF interface view

Default Level

2: Configuration level

Usage Guidelines

Usage Scenario

If a VLAN Down event occurs when all the interfaces added to the VLAN go Down, the VLAN will report the Down event to the VLANIF interface, causing a change in the VLANIF interface status. The master/slave switchover will cause the VLANIF interface to alternate frequently between Up and Down. As a result, network flapping occurs. To avoid this, run the **damping time** command on the VLANIF interface to enable VLAN damping. Among all the interfaces that are added to the VLAN, if the last Up interface in the VLAN becomes Down, the device enabled with VLAN damping will report the VLAN status to the VLANIF interface before the configured delay time expires. If some interfaces in the VLAN become Up before the configured delay time expires, the VLANIF interface stays Up.

Example

```
# Configure a device to report the VLAN 2 Down event to VLANIF 2 10 seconds  
later after all interfaces in VLAN 2 go Down.  
<HUAWEI> system-view  
[~HUAWEI] vlan 2  
[*HUAWEI-vlan2] quit  
[*HUAWEI] interface vlanif 2  
[*HUAWEI-Vlanif2] damping time 10
```

2.4.3.3 description (VLAN view)

Function

The **description** command is used to configure the description of a VLAN.

The **undo description** command is used to clear the description of a VLAN.

By default, the description of a VLAN contains the VLAN ID. For example, the description of VLAN 2 is VLAN 0002.

Format

description *description*

undo description

Parameters

Parameter	Description	Value
<i>description</i>	Indicates the description of a VLAN.	The value is a string of 1 to 80 characters, spaces supported.

Views

VLAN range view,VLAN view

Default Level

2: Configuration level

Usage Guidelines

Usage Scenario

To facilitate VLAN management and identification, run the **description** command to configure a description for a VLAN.

After this command is configured, run the **display vlan vlan-id verbose** command to view information about the VLAN.

Example

```
# Configure a description for VLAN as huawei.  
<HUAWEI> system-view  
[~HUAWEI] vlan 2  
[*HUAWEI-vlan2] description huawei
```